AI for Genomics: from CNNs and LSTMs to Transformers

Physalia course, online, 7-9 April 2026

Course outline and practical information

Nikolay Oskolkov, Group Leader (PI) at LIOS - NIRI, Riga, Latvia

Physalia Courses

About us

Organizer: Carlo Pecoraro, Physalia courses

info@physalia-courses.org

Instructor:

Dr. Nikolay Oskolkov, metabolic research group leader, LIOS

nikolay.oskolkov@osi.lv

@NikolayOskolkov

@oskolkov.bsky.social

Personal homepage:
https://nikolay-oskolkov.com

Brief introduction: who am I

Latvian Institute of Organic Synthesis

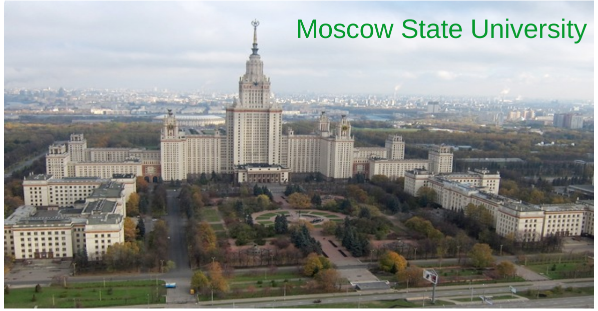
2007 PhD in theoretical physics in Moscow, Russia

2011 medical genomics at Lund University, Sweden

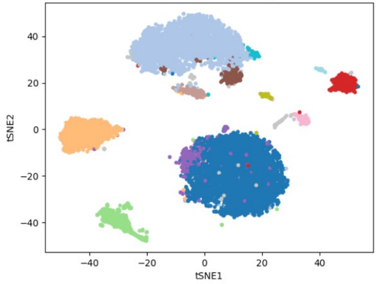
2016 bioinformatician at NBIS SciLifeLab, Sweden

2025 Metabolic Research Group leader, LIOS, Latvia

50+ publications; h-index = 25; 4,000+ citations



Single cell



Biomedical data integration

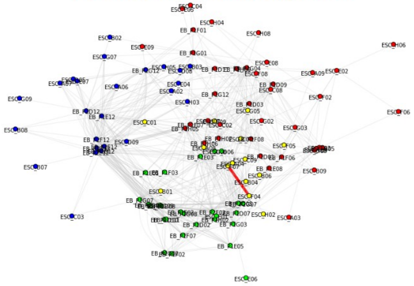
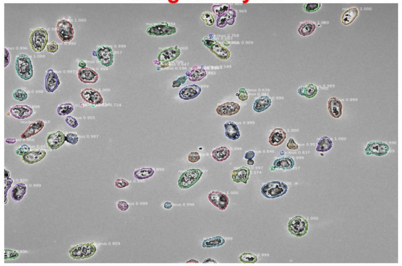
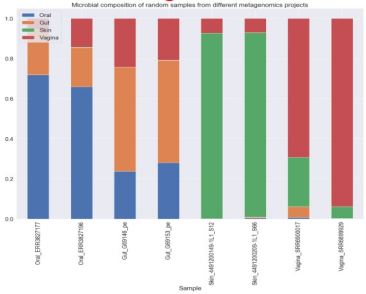


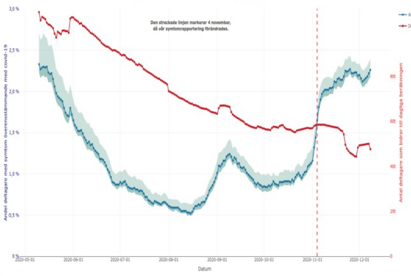
Image analysis



Metagenomics



Epidemiology



Ancient microbiome

Metabolic Research Group at LIOS - NIRI

Latvian Institute of Organic Synthesis

PhD. Nikolay Oskolkov
Group Leader (PI) of the Metabolic Research Group

1 more postdoctoral fellow and 1 PhD student to be hired

If you know anyone who might be interested, please contact me!

Kristina Grausa, PhD sudent
in Metabolic Research Group

Daniel Rivas, MD, PhD in AI,
postdoctoral fellow
in Metabolic Research Group

About you

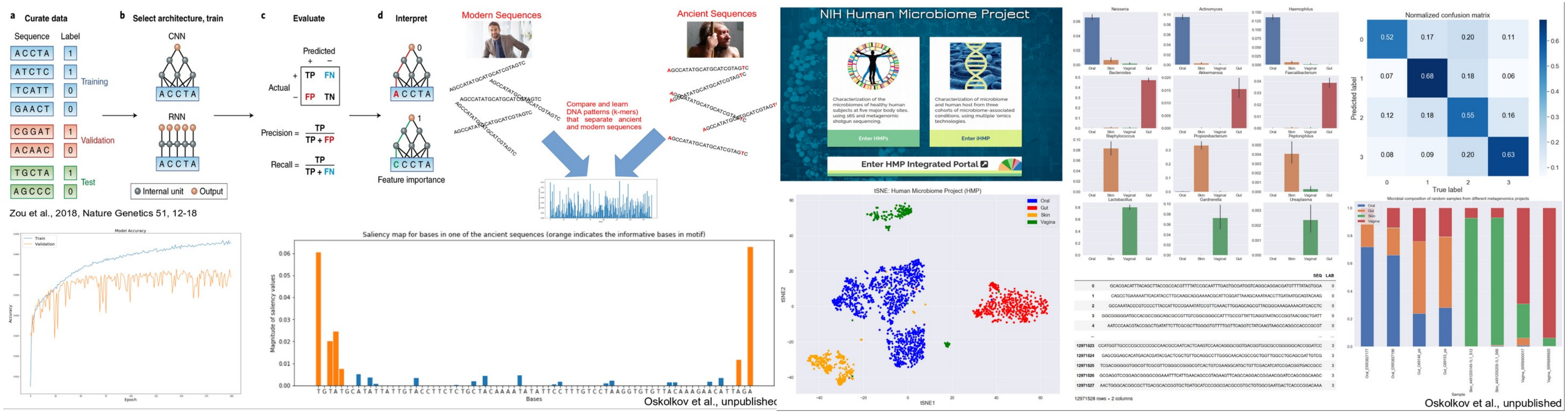
- Name
- University / Institute / Company
- Research interest(s)
- Previous experience with computational analysis and bioinformatics
- Motivation to join the course
- Expectations from the course

Course outline

Day 1: introduction to machine and deep learning, applications of ANNs in Life Sciences, coding gradient descent and a vanilla ANN from scratch in R and Python

Day2: CNNs for microscopy imaging: object detection and segmentation, autoencoder for single cell genomics and integration of heterogeneous data

Day3: convolutional (CNN) and recurrent (LSTM) neural networks, their applications in genomics and metagenomics, attention mechanism and Transformer



Housekeeping rules

- To ask question please raise your hand, unmute yourself and ask. You can also ask questions in the zoom chat or slack channel for this seminar.

- Please keep your camera on as much as possible for better contact and communication.

- The course includes 8 lectures (~1h each) and 11 practicals (~1.5h each), there is a 15-30 min break after each lecture and each practical

- During practicals, we will use Rmarkdown and Jupyter notebooks. At the end of each practical, I will use html-versions of the notebooks to go through command lines with my explanations.

- The material is based on data and problems from computational biology, however the concepts discussed are general and can be applied for other types of data.

Practical information: GitHub and Zoom

The course will take place in Zoom from 2 pm to 8 pm (CET, Berlin time)

Links to the Zoom room will be posted in the Slack channel

The course GitHub repository containing lectures and exercises is:

https://github.com/NikolayOskolkov/Physalia_AI_Genomics

Please bookmark this address!

Physalia Courses

Physalia_AI_Genomics Public

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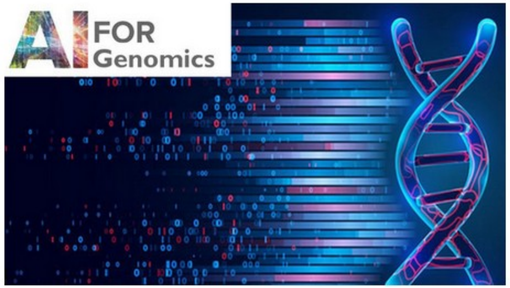
main | 1 Branch | 0 Tags | Go to file | Add file | Code

LeandroRitter edited README c1670c4 · 14 hours ago 22 Commits

practicals	added material for Day3	2 days ago
slides	added material for Day3	2 days ago
AI_Genomics_Oskolkov_ syllabus.pdf	added logo and modified the program	3 days ago
README.md	edited README	14 hours ago
command-line-basics.md	added logo and modified the program	3 days ago
course_logo.jpg	added logo and modified the program	3 days ago

README



AI for Genomics

Instructor

- Dr. Nikolay Oskolkov, Group Leader (PI) at LIOS, Riga, Lativia

Course overview

This course explores the application of modern AI architectures—Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and Transformers—to genomic and metagenomic data. Students will gain practical experience through hands-on coding labs and interactive notebooks, learning how to model sequence data, extract biologically meaningful features, and interpret results. Emphasis is placed on real-world applications, including prediction of genomic functional elements, sequence classification and source

About

This is a repository with the course material for Physalia AI for Genomics course

Readme
Activity
1 star
0 watching
0 forks

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 2

LeandroRitter Nikolay Oskolkov
NikolayOskolkov Nikolay Oskolkov

Languages

- HTML 51.6%
- Jupyter Notebook 48.4%

Suggested workflows

Based on your tech stack

Jekyll using Docker image Configure
Package a Jekyll site using the jekyll/builder Docker image.

SLSA Generic generator Configure
Generate SLSA3 provenance for your existing release workflows

More workflows Dismiss suggestions

Practical Information: we will be using Jupyter and Rmarkdown notebooks

To run Jupyter notebooks on Google Colab please do the following:

1. Clone this repo: https://github.com/NikolayOskolkov/Physalia_AI_Genomics
2. Start the Jupyter notebook from the Google drive and mount google drive:

```
#Mount Google Drive to Colab

from google.colab import drive
drive.mount('/content/drive')

!ls -l /content/drive/MyDrive/

import os
os.chdir("/content/drive/MyDrive/data")
```

N.B. you need a Google account to be able to run Google Colab!

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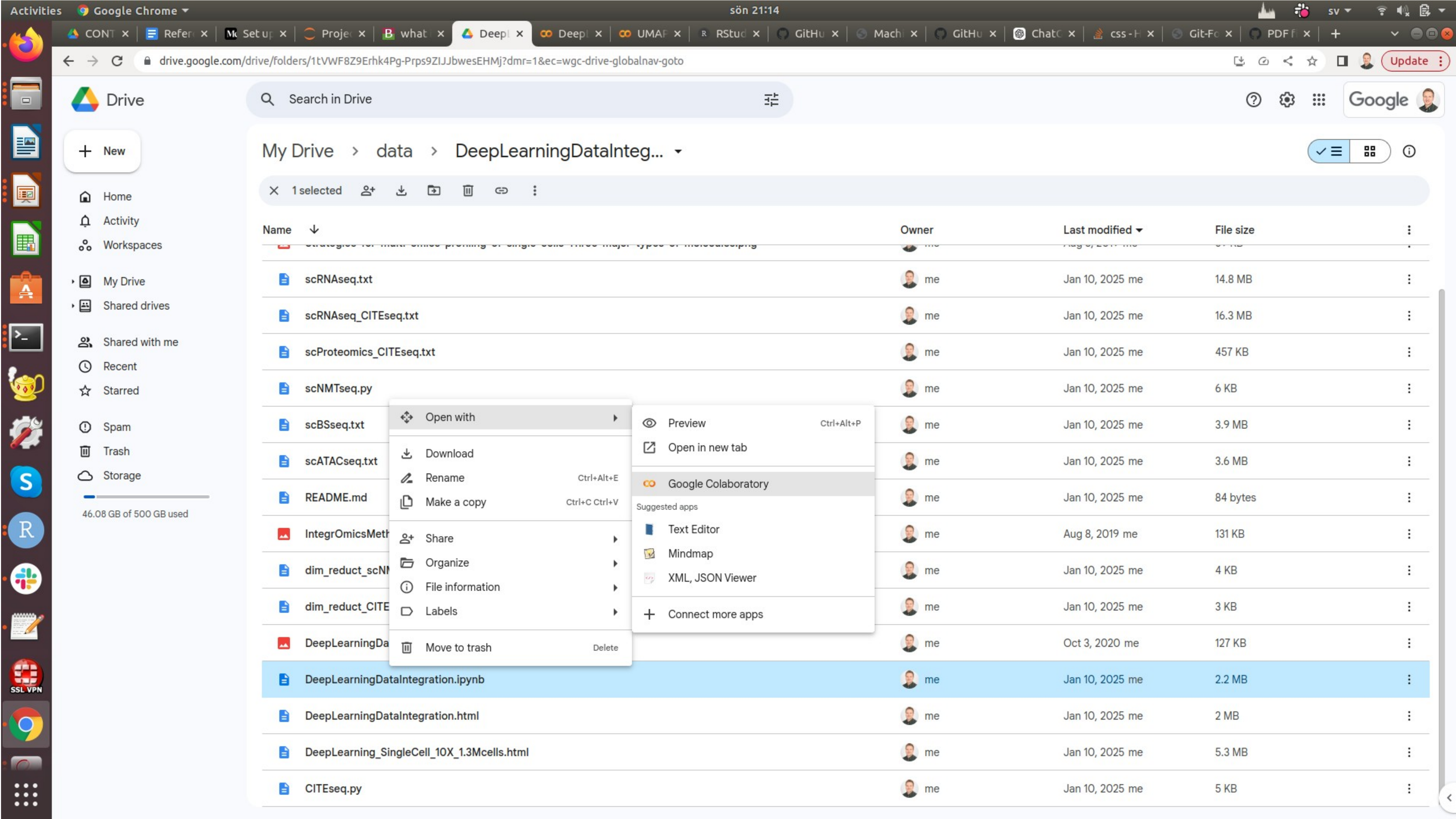
drive.google.com/drive/folders/1tVWF8Z9Erhk4Pg-Prps9ZIJJbwesEHMj?dmr=1&ec=wgc-drive-globalnav-goto
Drive
Search in Drive
New
Home
Activity
Workspaces
My Drive
Shared drives
Shared with me
Recent
Starred
Spam
Trash
Storage
46.08 GB of 500 GB used
My Drive > data > DeepLearningDataInteg...
1 selected
Name
Owner
Last modified
File size
scRNAseq.txt me Jan 10, 2025 me 14.8 MB
scRNAseq_CITEseq.txt me Jan 10, 2025 me 16.3 MB
scProteomics_CITEseq.txt me Jan 10, 2025 me 457 KB
scNMTseq.py me Jan 10, 2025 me 6 KB
scBSseq.txt me Jan 10, 2025 me 3.9 MB
scATACseq.txt me Jan 10, 2025 me 3.6 MB
README.md me Jan 10, 2025 me 84 bytes
IntegrOmicsMeth me Aug 8, 2019 me 131 KB
dim_reduct_scNI me Jan 10, 2025 me 4 KB
dim_reduct_CITE me Jan 10, 2025 me 3 KB
DeepLearningDa me Oct 3, 2020 me 127 KB
DeepLearningDataIntegration.ipynb me Jan 10, 2025 me 2.2 MB
DeepLearningDataIntegration.html me Jan 10, 2025 me 2 MB
DeepLearning_SingleCell_10X_1.3Mcells.html me Jan 10, 2025 me 5.3 MB
CITEseq.py me Jan 10, 2025 me 5 KB
Open with
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Rename Ctrl+Alt+E
Make a copy Ctrl+C Ctrl+V
Share
Organize
File information
Labels
Move to trash Delete
Preview Ctrl+Alt+P
Open in new tab
Google Colaboratory
Suggested apps
Text Editor
Mindmap
XML, JSON Viewer
Connect more apps

Deep Learning for Data Integration

Biological and biomedical research has been tremendously benefiting last decade from the technological progress delivering DNA sequence (genomics), gene expression (transcriptomics), protein abundance (proteomics) and many other levels of biological information commonly referred to as OMICs. Despite individual OMICs layers are capable of answering many important biological questions, their combination and consequent synergistic effects from their complementarity promise new insights into behavior of biological systems such as cells, tissues and organisms. Therefore OMICs integration represents the contemporary challenge in Biology and Biomedicine.

```
[ ] from google.colab import drive
    drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
[ ] !ls -l /content/drive/My\ Drive/data/DeepLearningDataIntegration
```

```
total 49588
-rw------- 1 root root     5105 Jan 10 12:26 CITEseq.py
-rw------- 1 root root  2066933 Jan 10 12:26 DeepLearningDataIntegration.html
-rw------- 1 root root  1747228 Jan 10 12:41 DeepLearningDataIntegration.ipynb
-rw------- 1 root root   130233 Oct  3  2020 DeepLearningDataIntegration.jpg
-rw------- 1 root root  5598515 Jan 10 12:26 DeepLearning_SingleCell_10X_1.3Mcells.html
-rw------- 1 root root     3499 Jan 10 12:26 dim_reduct_CITEseq.py
-rw------- 1 root root     3616 Jan 10 12:26 dim_reduct_scNMTseq.py
-rw------- 1 root root   134229 Aug  8  2019 IntegrOmicsMethods.png
-rw------- 1 root root       84 Jan 10 12:26 README.md
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-rw------- 1 root root  4082712 Jan 10 12:26 scBSseq.txt
-rw------- 1 root root     6141 Jan 10 12:26 scNMTseq.py
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-rw------- 1 root root 17128751 Jan 10 12:26 scRNAseq_CITEseq.txt
-rw------- 1 root root 15547715 Jan 10 12:26 scRNAseq.txt
-rw------- 1 root root    71008 Aug  8  2019 Strategies-for-multi-omics-profiling-of-single-cells-Three-major-types-of-molecules.png
-rw------- 1 root root     3647 Jan 10 12:26 tSNE_on_Autoencoder_CITEseq.py
-rw------- 1 root root     3635 Jan 10 12:26 tsne_on_autoencoder_scNMTseq.py
```

```
import os
os.chdir("/content/drive/My Drive/data/DeepLearningDataIntegration")
```

```
[ ] from IPython.display import Image
    Image('DeepLearningDataIntegration.jpg', width=2000)
```